Kinematics

Kinematics

- Kinematics is a part of mechanics that enables us to describe motion.
- We need to define some physical quantities which can help describe motion in one, two and three dimensions.

Motion in one dimension

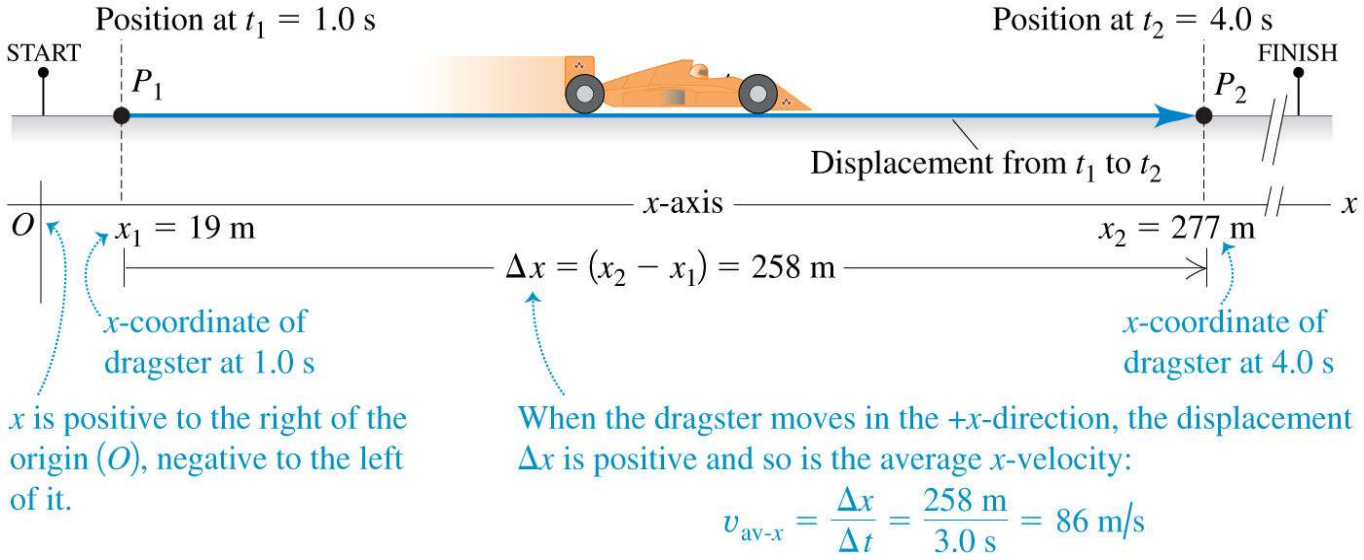
- The simplest kind of motion is motion in a straight line.
- Examples of this kind of motion include an object moving on a straight road and a stone thrown vertically upward.
- We will assume that the objects are represented by point particles instead of considering them as extended objects.

Motion of a particle in a straight line

- To describe the motion of a particle, we need a coordinate system.
- We choose the x-axis with an origin O to lie along the road.
- The change in the particle's coordinate *x* over a time interval will describe the motion of the particle over that time interval.

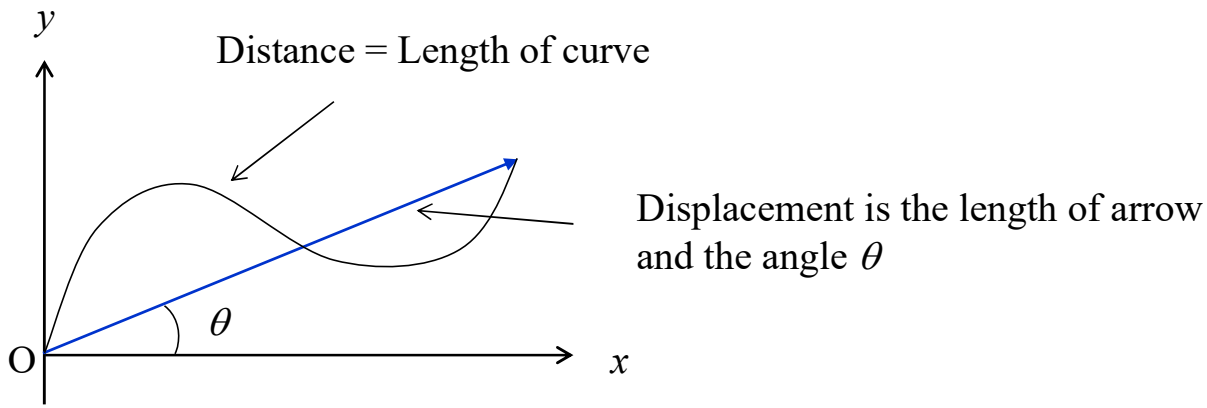
Displacement

- The displacement of the particle is a vector pointing from P_1 to P_2 (see below figure).
- The x-component of the displacement is given by the change in the particle's coordinate i.e. $\Delta x = x_2 - x_1$.



Distance vs displacement

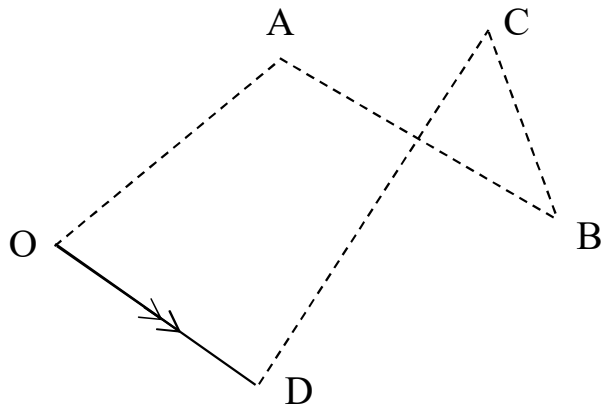
- Distance is the length of the path travelled by an object.
- Displacement is the straight line length from the start point to the end point plus the direction of that line with respect to a reference direction.
- Distance is a scalar and displacement is a vector.



Example 1

Smoke particle undergoes Brownian motion from O to A to B to C to D.

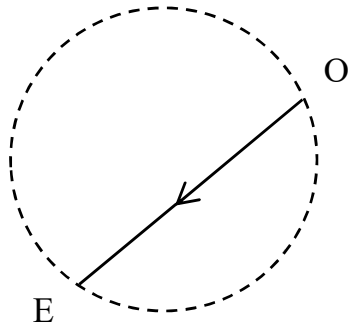
a) What is the distance travelled to D from O?
b) What is the displacement of D from O?



Example 2

A body moves in a circle in the clockwise direction starting from O to E.

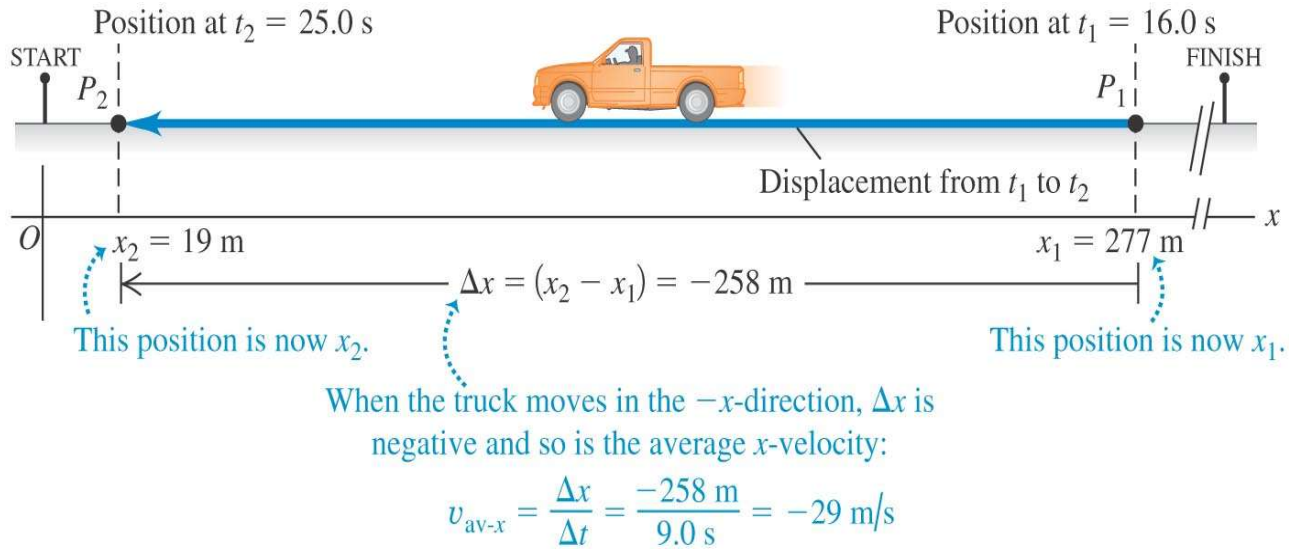
a) What is the distance travelled by the body from O to E?
b) What is the displacement of the body at E from O?
c) What is the distance travelled by the body when it returns to O?
d) What is the displacement of the body when it returns to O?



Average velocity

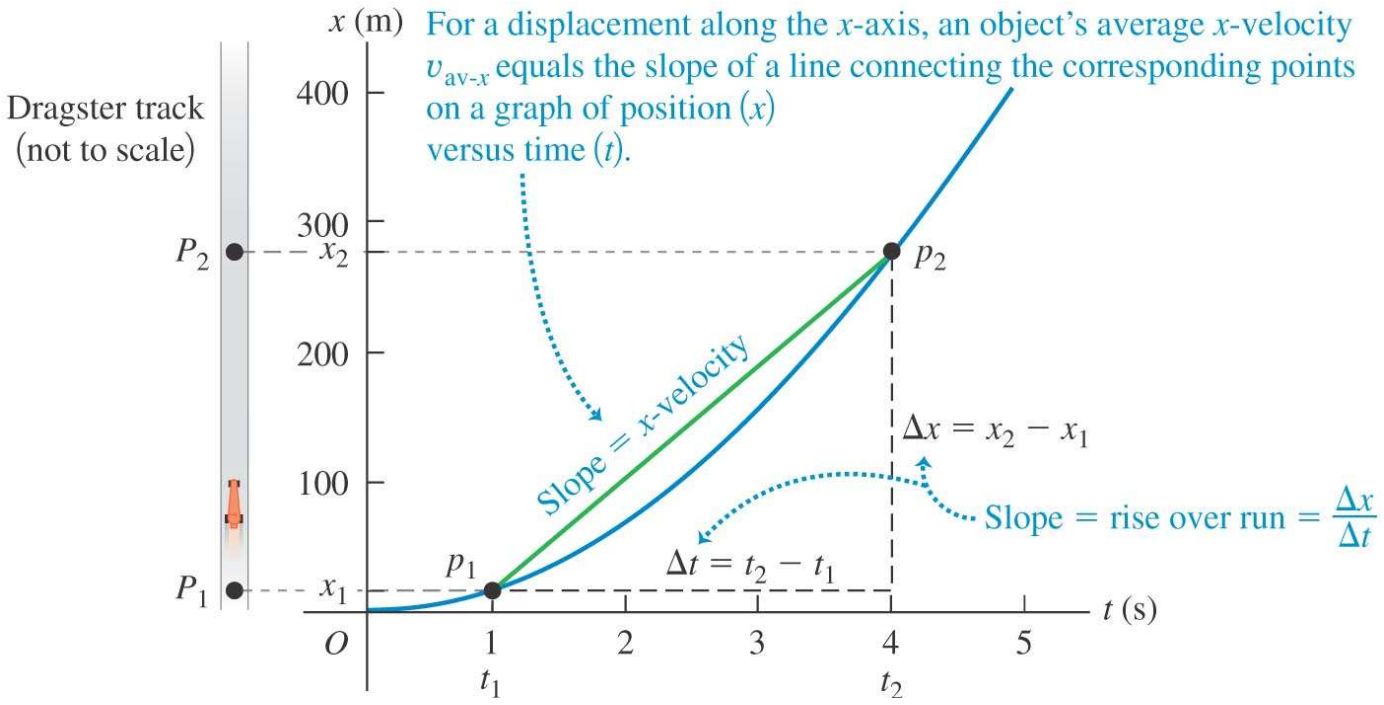
- The average velocity is a vector whose x-component is defined as

$$v_{av,x} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\Delta x}{\Delta t}$$



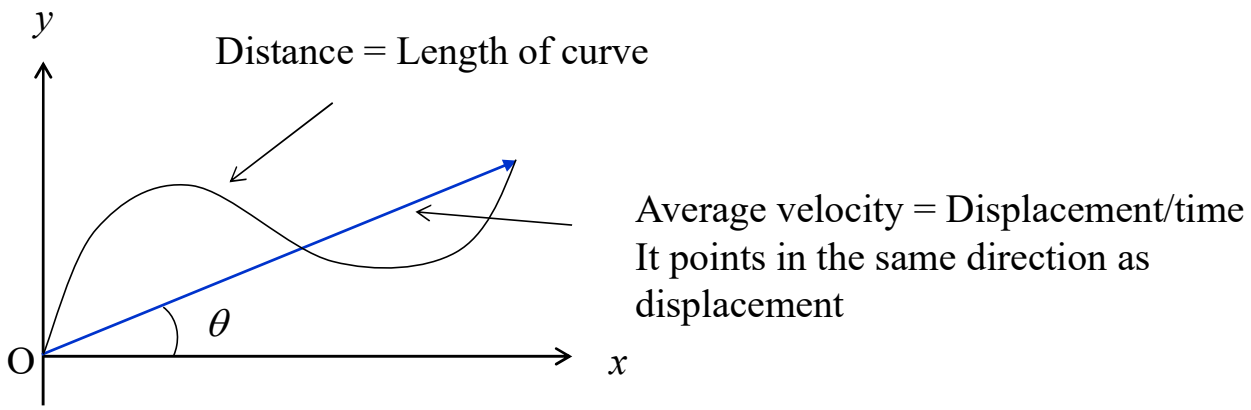
Position-time graph *x*-*t* graph

- The average *x*-velocity $v_{av,x}$ is the slope of an *x*-*t* graph.



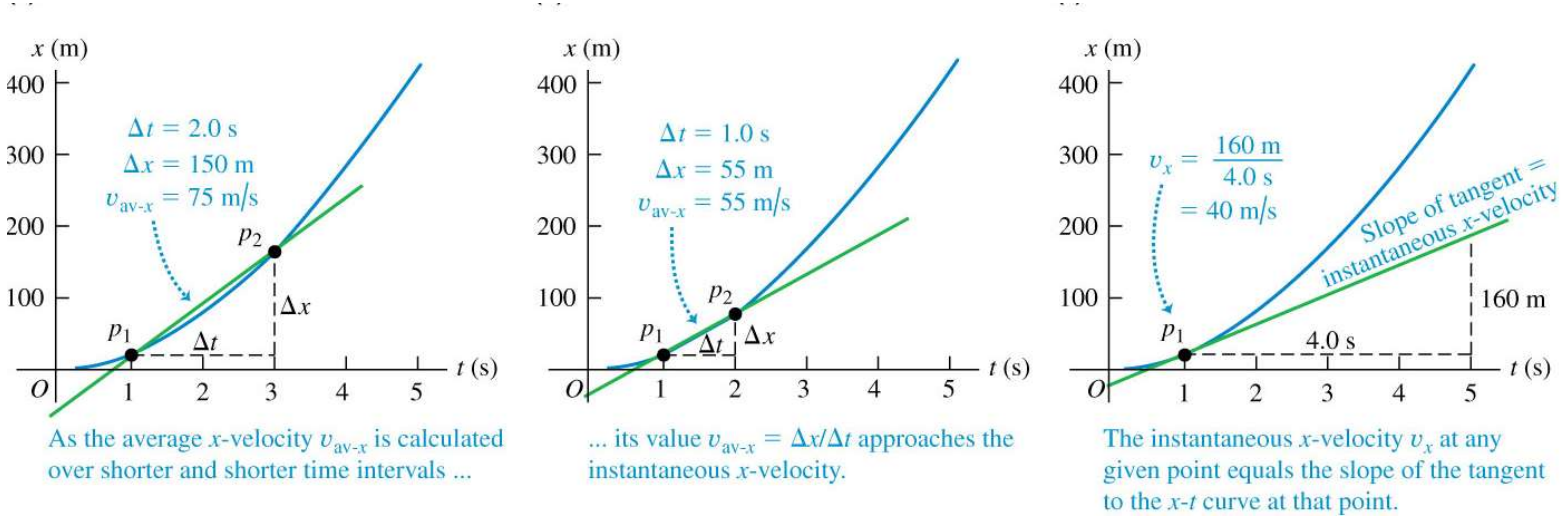
Speed and velocity

- Average speed is distance divided by the time taken.
- Average velocity is the displacement divided by time taken.
- The average speed is greater than or equal to the magnitude of the average velocity.



Instantaneous velocity for straight line motion

- The average velocity of a particle during a time interval can't tell us how fast or in what direction the particle was moving at any given time during the interval.
- Therefore we introduce the concept of instantaneous velocity which is the velocity at a specific instant of time or specific point along the path.



As the average x-velocity $v_{av\text{-}x}$ is calculated over shorter and shorter time intervals ...

... its value $v_{av\text{-}x} = \Delta x/\Delta t$ approaches the instantaneous x-velocity.

The instantaneous x-velocity v_x at any given point equals the slope of the tangent to the x-t curve at that point.

Instantaneous velocity for straight line motion

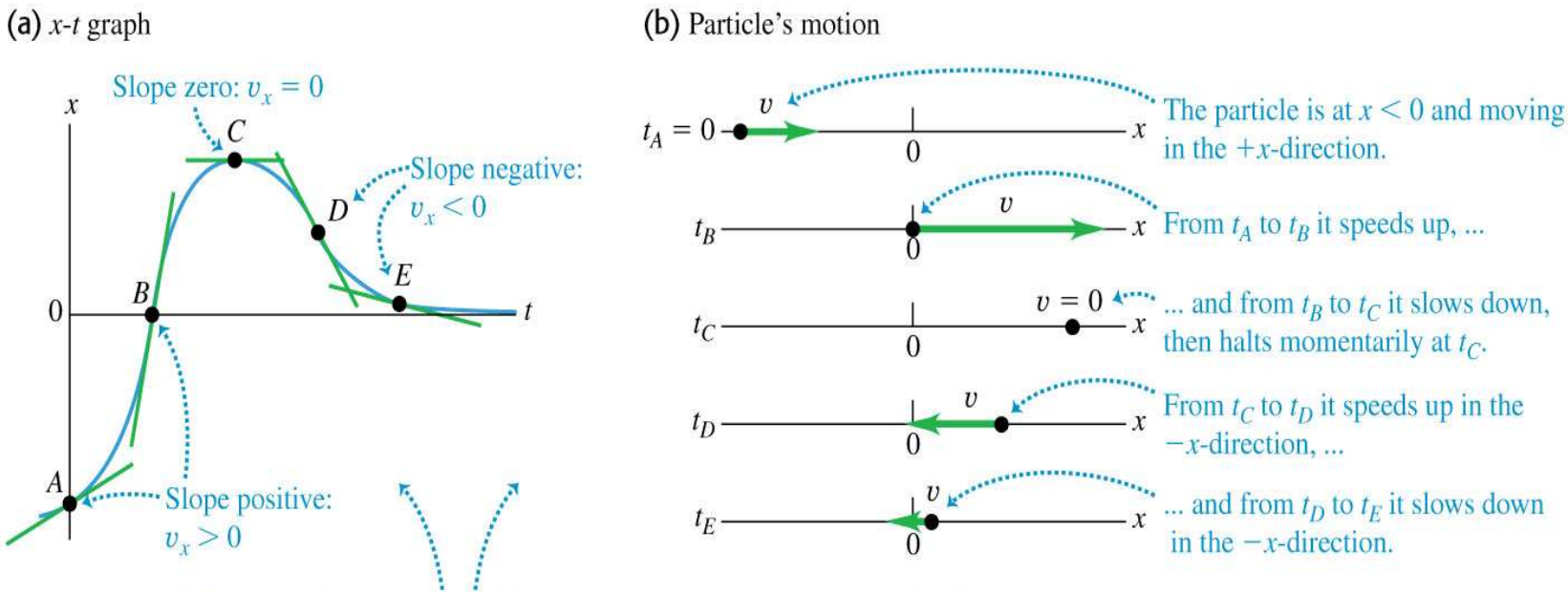
- The instantaneous velocity is the limit of the average velocity as the time interval approaches zero.

$$\vec{v}_x = \lim_{\Delta t \to 0} \frac{\Delta \vec{x}}{\Delta t} = \frac{d\vec{x}}{dt}$$

- Unless specified average velocity, you can assume that the term velocity implies instantaneous velocity.
- The SI unit of velocity is m/s.

Instantaneous velocity from *x*-*t* graph for straight line motion

- The instantaneous velocity at any point on *x*-*t* graph is equal to the slope of the tangent at that point.



The steeper the slope (positive or negative) of an object's *x*-*t* graph, the greater is the object's speed in the positive or negative *x*-direction.

Key words in a problem

Key

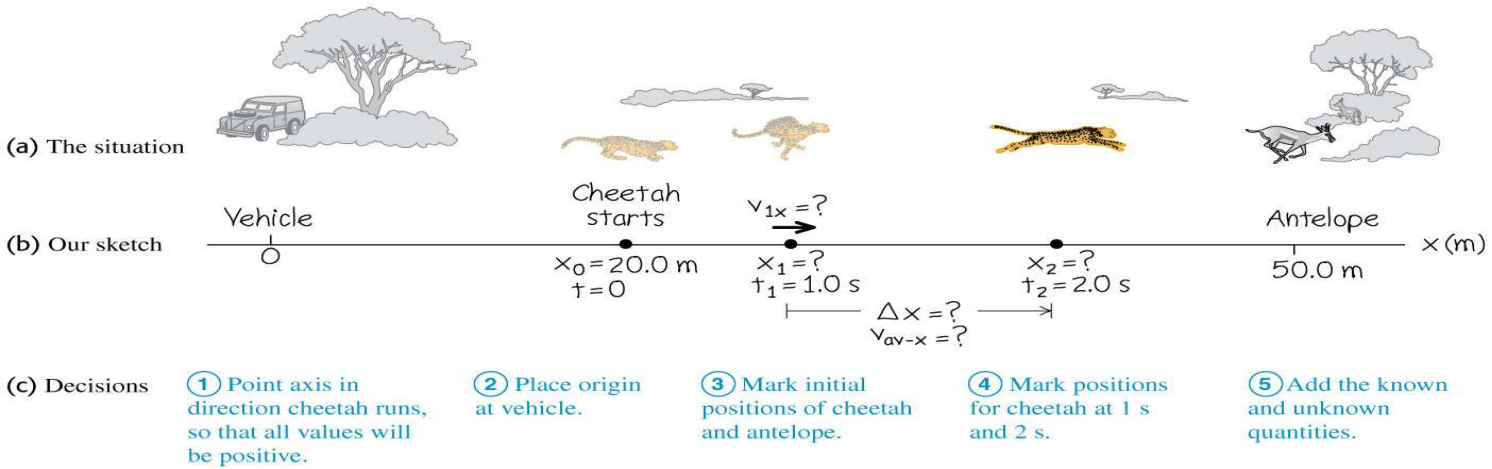
1. When → Calculate time
2. Initial → at $t = 0$
3. Rest → velocity $= 0$
4. Where → position / distance
5. change in x → $\Delta x = (x_f) - (x_i)$ (final value − initial value)
6. Compare → ratio : A/B
7. Difference → $x_A - x_B$

Show or Prove ⇒ → LHS. = RHS.

Example 3

A cheetah is crouched 20 m to the east of an observer sitting in a vehicle (see below figure). At $t = 0$ the cheetah begins to run due eastward toward an antelope that is 50 m to the east of the observer. During the first 2.0 s of the attack, the cheetah's coordinate x varies with time according to $x = 20\text{ m} + (5.0\text{ m/s}^2)t^2$

a) Find the cheetah's displacement between $t_1 = 1.0$ s and $t_2 = 2.0$ s.
b) Find the average velocity during that interval.
c) Derive an expression for the cheetah's instantaneous velocity as a function of time and use it to find v_x at $t = 1.0$ s and $t = 2.0$ s.



Average acceleration for straight line motion

- The average acceleration, $a_{av,x}$ of the particle is a vector whose x-component is defined as the change in x-component of the velocity divided by the time interval. That is,

$$\vec{a}_{av,x} = \frac{\vec{v}_{2x} - \vec{v}_{1x}}{t_2 - t_1} = \frac{\Delta \vec{v}_x}{\Delta t}$$

- The SI unit of average acceleration is m/s^2.

Instantaneous acceleration for straight line motion

- The instantaneous acceleration is the rate of change of velocity with time and is given by :

$$\vec{a}_x = \lim_{\Delta t \to 0} \frac{\Delta \vec{v}_x}{\Delta t} = \frac{d\vec{v}_x}{dt}$$

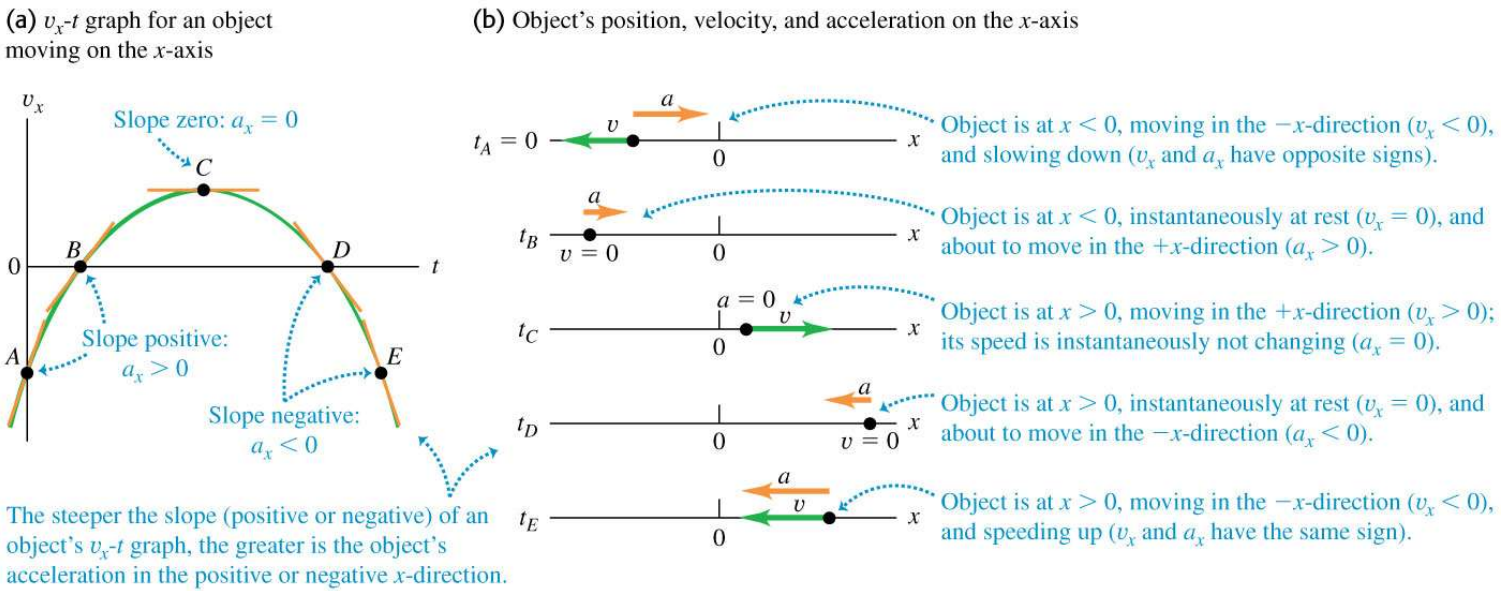
$$= \frac{d^2}{dt^2}\vec{x}$$

- Unless specified average acceleration, you can assume that the term acceleration implies instantaneous acceleration.

- The SI unit of acceleration is (m/s/s) or simply m/s^2.

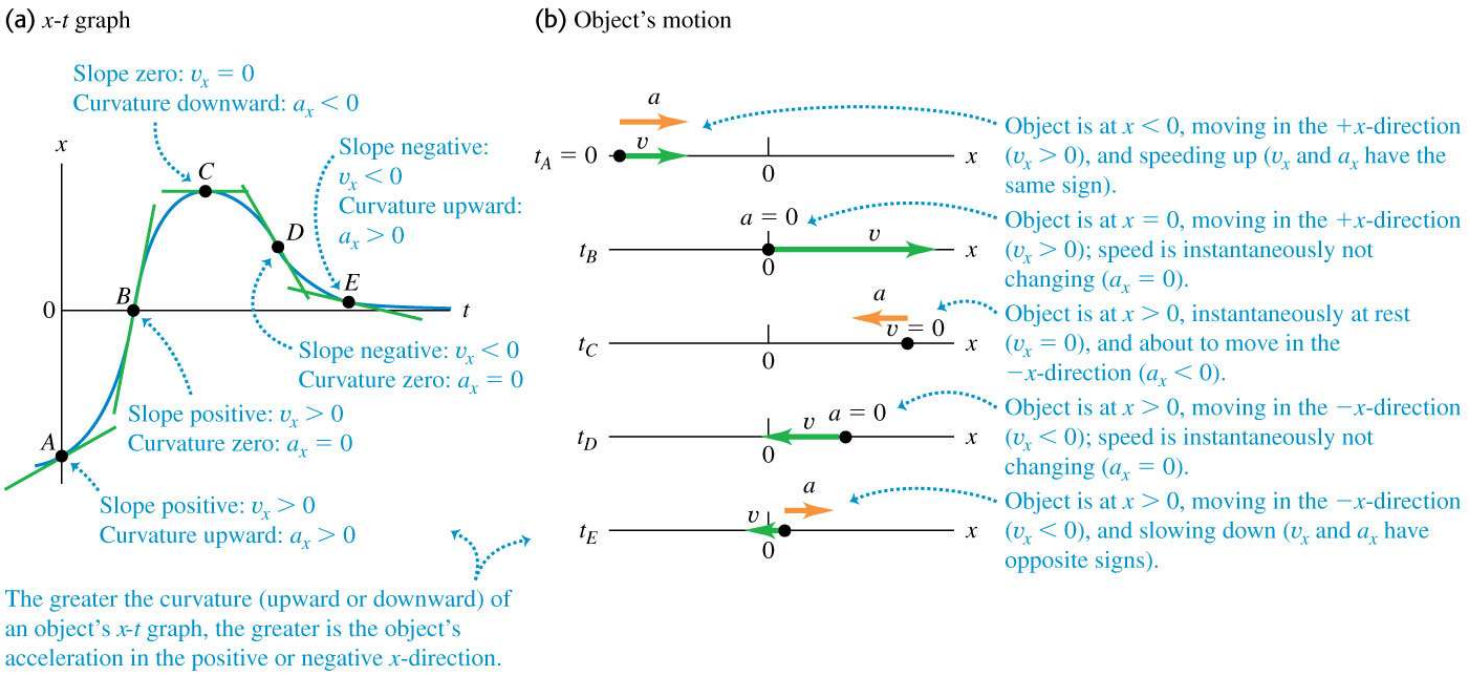
Instantaneous acceleration from v_x-t graph for straight line motion

- The instantaneous acceleration at any point on the v-t graph is equal to the slope of the tangent to the curve at that point.



Instantaneous acceleration from *x*-*t* graph for straight line motion

- On an *x*-*t* graph, the instantaneous acceleration at any point is given by the concavity or curvature of the graph at that point.



Rules for the sign of x-acceleration

- Deceleration means a decrease in speed. Negative acceleration does not always mean a decrease in speed.

If x-velocity is:	**. . . x-acceleration is:**
Positive & increasing (getting more positive)	Positive: Particle is moving in $+x$-direction & speeding up
Positive & decreasing (getting less positive)	Negative: Particle is moving in $+x$-direction & slowing down
Negative & increasing (getting less negative)	Positive: Particle is moving in $-x$-direction & slowing down
Negative & decreasing (getting more negative)	Negative: Particle is moving in $-x$-direction & speeding up

Note: These rules apply to both the average x-acceleration $a_{\text{av-}x}$ and the instantaneous x-acceleration a_x.

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Example 4

The x-velocity v_x, of a car at time t is given by the equation $v_x = 60\ \text{m/s} + (0.50\ \text{m/s}^3)t^2$

a) Find the change in x-velocity of the car in the time interval $t_1 = 1.0$ s and $t_2 = 3.0$ s.

b) Find the average x-acceleration during this time interval.

c) Derive an expression for the instantaneous acceleration as a function of time and use it to find a_x at $t = 1.0$ s and $t = 3.0$ s.

Straight line motion with constant acceleration

- Let a particle have a velocity v_{0x} at time $t = 0$ which we call the initial velocity and v_x at a later time t which we call the final velocity.
- From the definition of average acceleration we get

$$a_{av-x} = a_x = \frac{v_x - v_{0x}}{t - 0}$$

- Rearranging the above to get the relation gives

$$v_x = v_{0x} + a_x t$$

Straight line motion with constant acceleration

- If the position of the particle at $t = 0$ is x_0 and moves to position x at time t, then the average velocity is

$$v_{av-x} = \frac{x - x_0}{t - 0}$$

- We also know that the average velocity can be written as

$$v_{av-x} = \frac{(v_{0x} + v_x)}{2}$$

- The above two equations can be combined to yield

$$x - x_0 = \frac{(v_{0x} + v_x)t}{2}$$

Straight line motion with constant acceleration

- Simplifying we get a useful equation that relates the final position with the initial position, initial velocity, acceleration and time.

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$$

- We can eliminate time using $t = \frac{v_x - v_{0x}}{a_x}$ and on substituting in the above equation we get

$$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$$

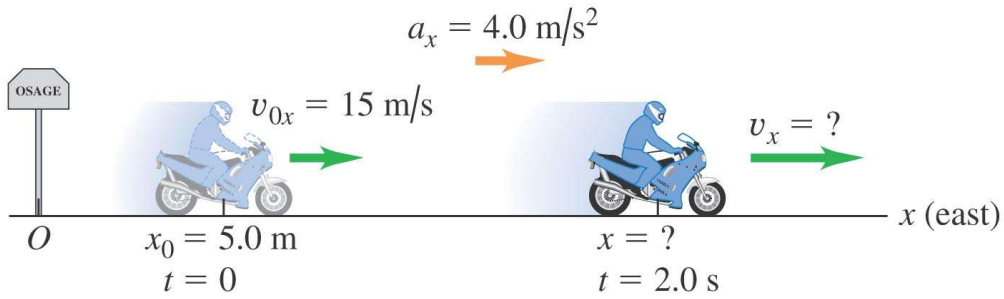
- Note that the equations we have derived are applicable only when the acceleration is constant.

Example 5

A motorcyclist heading east through a small town accelerates at a constant 4.0 m/s^2 after he leaves the city limits (see below figure). At time $t = 0$ he is 5.0 m east of the city-limits signpost, moving east at 15 m/s.

a) Find his position and velocity at $t = 2.0$ s.

b) Where is he when his velocity is 25 m/s?

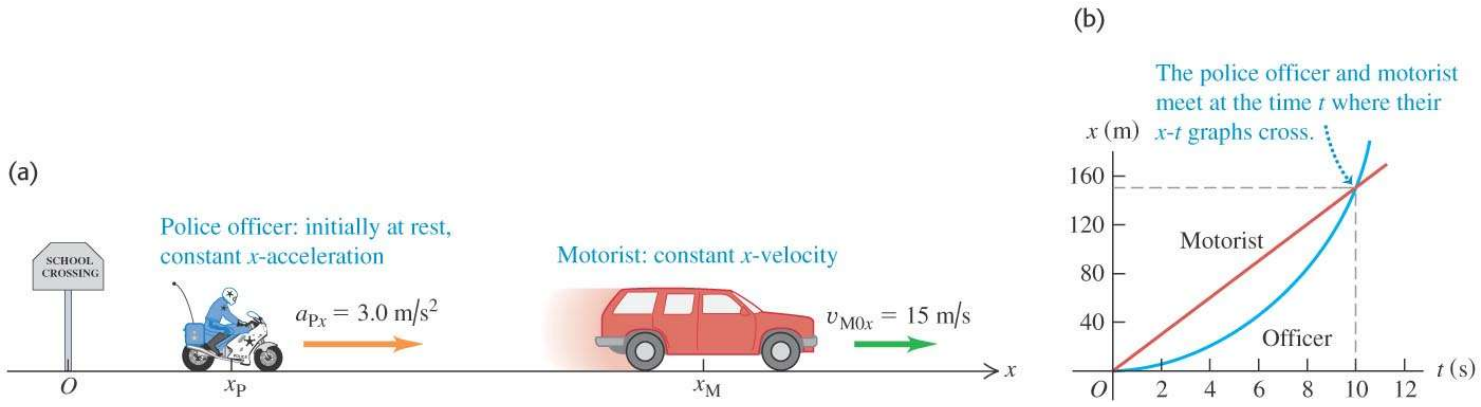


Example 6

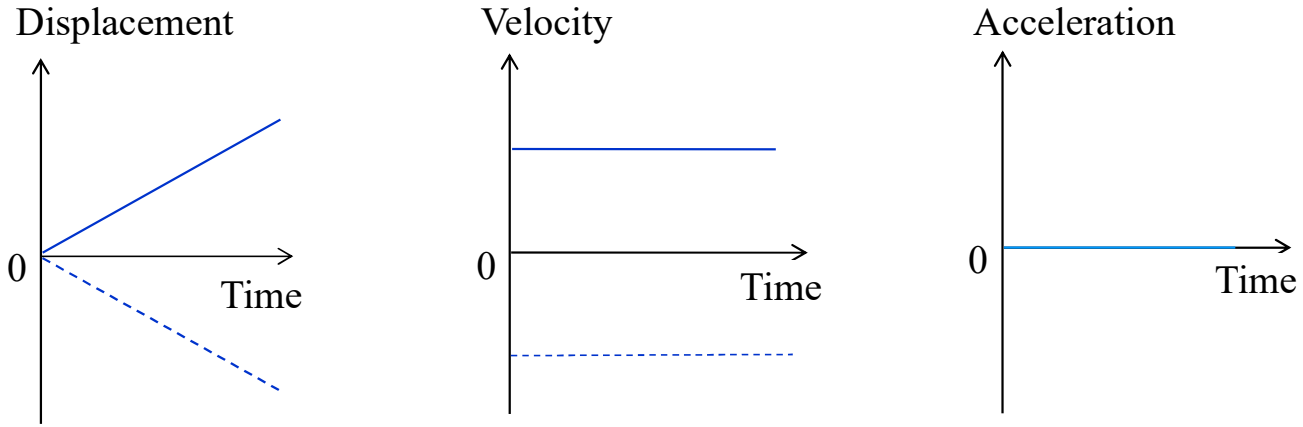
A motorist travelling with a constant speed of 15 m/s passes a school-crossing corner where the speed limit is 10 m/s. Just as the motorist passes the school-crossing sign, a police officer on a motorcycle, initially at rest, starts in pursuit with a constant acceleration of 3.0 m/s^2 (see below figure).

a) How much time elapses before the officer passes the motorist?

b) At that time, what distance has each vehicle travelled?



Graphs of uniform velocity



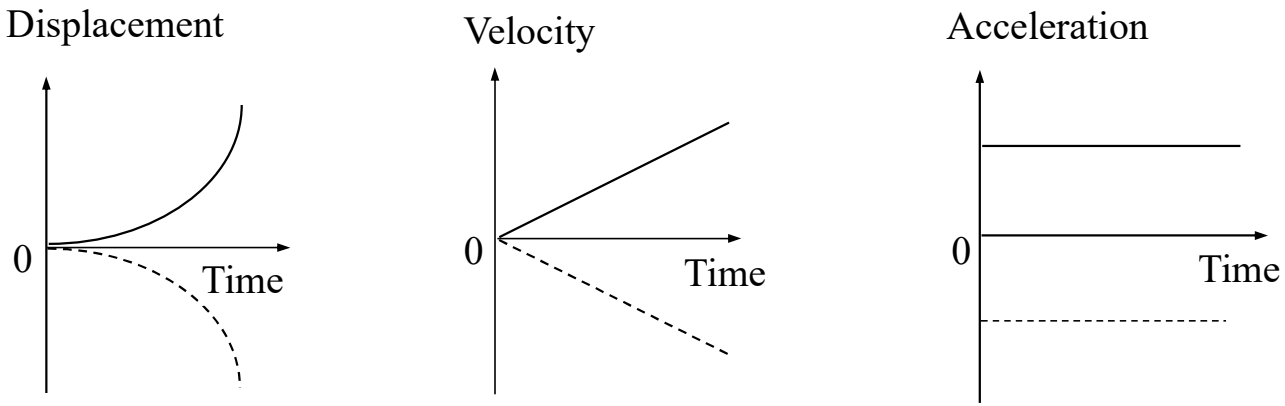
(a) (b) (c)

a) Displacement increases linearly with time.

b) Velocity remains the same with time.

c) Acceleration is zero.

Graphs of uniform acceleration



(a) (b) (c)

a) Displacement increases non-linearly with time.

b) Velocity increases linearly with time.

c) Acceleration is non zero and constant with time.

Free fall - one dimensional motion with constant acceleration

- Free fall motion is when an object falls under earth's gravity assuming that
  - there is no air resistance,
  - the distance of the fall is small compared to the radius of the earth,
  - the rotation of the earth is not important .
- The acceleration due to gravity is taken to be a constant irrespective of the size and mass of the object.
- The magnitude of the acceleration due to gravity on Earth's surface is $9.80\ m/s^2$.

Example 7

You throw a ball vertically upward from the roof of a tall building. The ball leaves your hand at a point even with the roof railing with an upward speed of 15.0 m/s. On its way back down, it just misses the railing.

a) Find the ball's position and velocity 1.00 s and 4.00 s after leaving your hand.

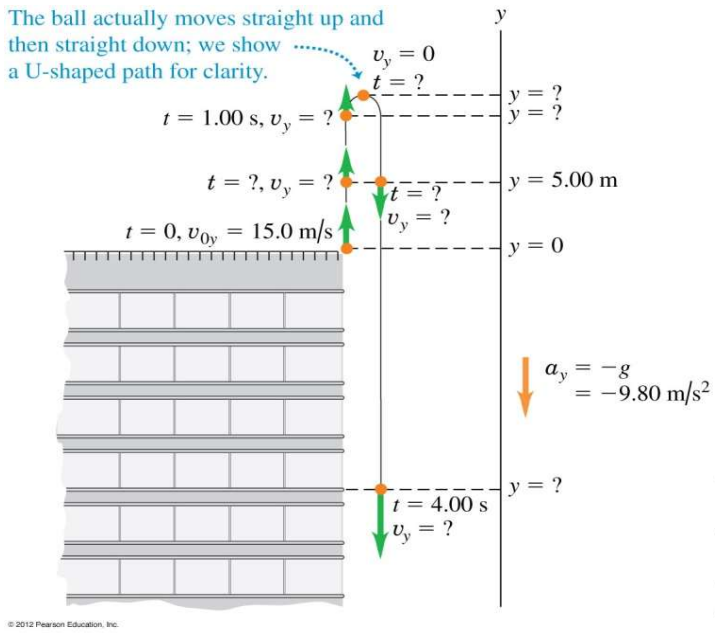
b) Find the ball's velocity when it is 5.00 m above the railing.

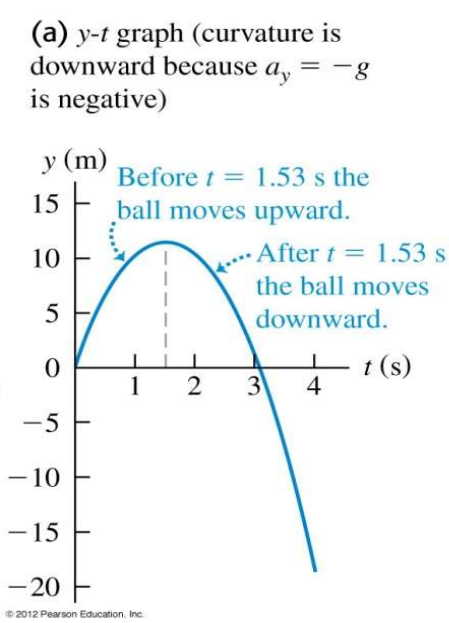
c) Find the maximum height reached.

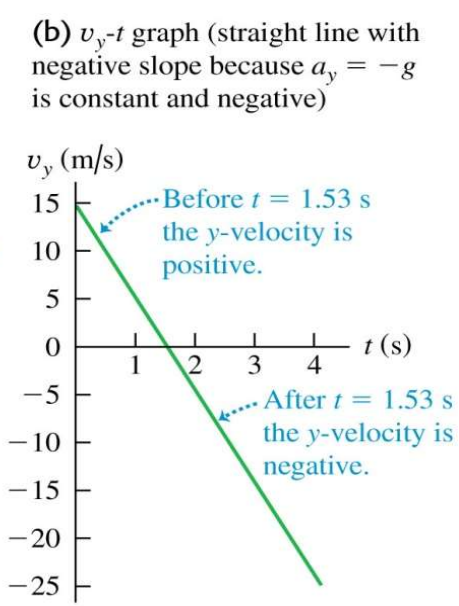
d) Find the ball's acceleration when it is at its maximum height.

e) At what time after being released has the ball fallen 5.00 m below the roof railing?

Example 7 - cont







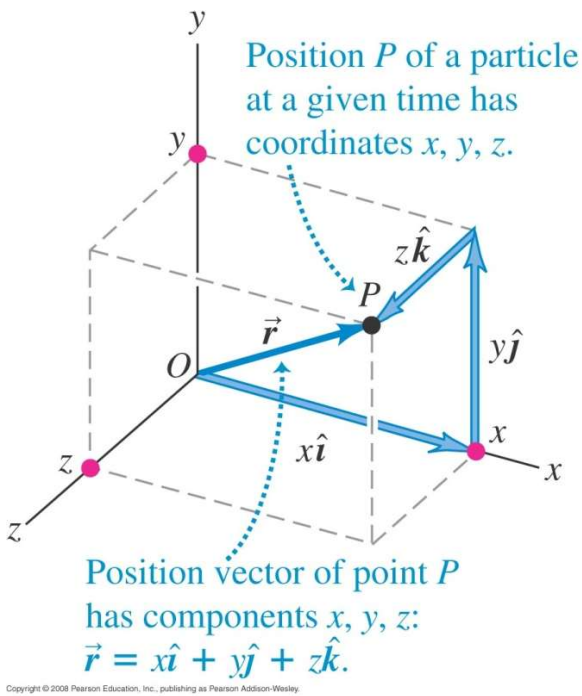
Motion in two and three dimensions

- Examples include a merry-go-round undergoes a circular motion, a base ball when hit may move in path that is not a straight line.
- The equations that we set up for studying motion in one dimension will be extended to two and three dimensions.

Position vector

- The position of a particle is a vector drawn from the origin of a co-ordinate system to where the particle is located at P is given by

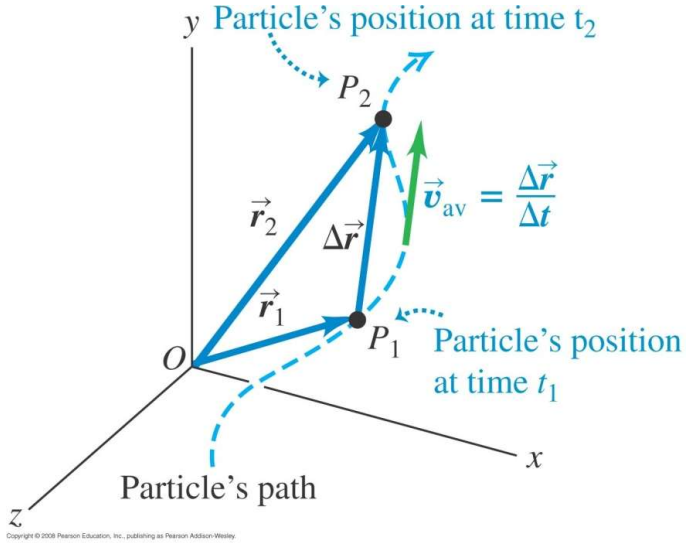
$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$



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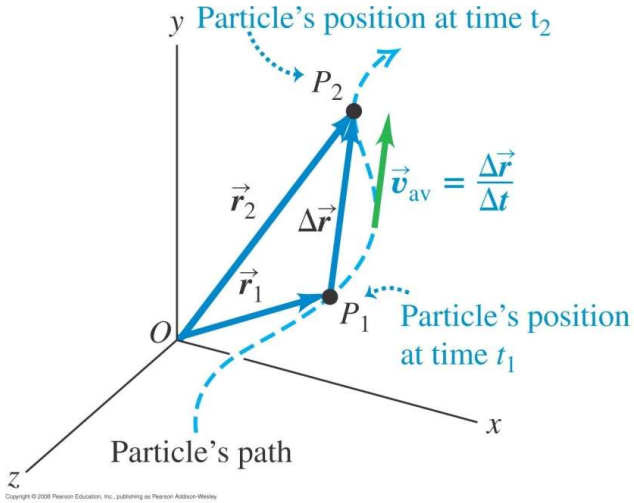
Displacement vector

- The displacement of a particle that moves from P_1 to P_2 is defined as $\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$
- For a curved path, the magnitude of the displacement vector is less than the distance travelled.
- Displacement vector is independent of the path, i.e. depends only on the start and end points.



Average velocity

- The average velocity is defined as $\vec{v}_{av} = \dfrac{\Delta \vec{r}}{\Delta t}$
- It has the same direction as the displacement vector.

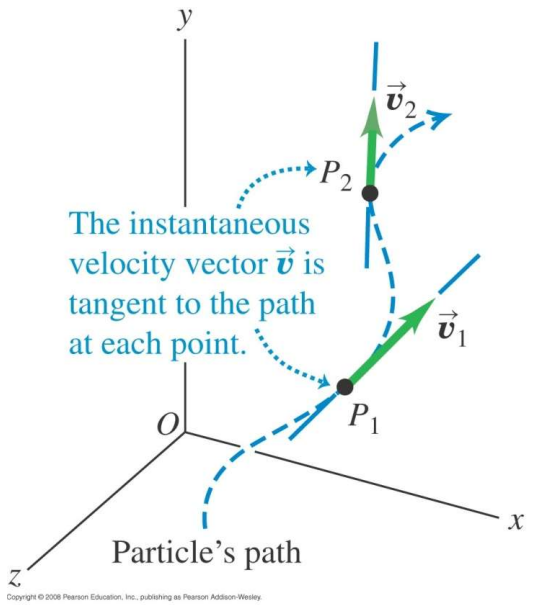


Instantaneous velocity

- The instantaneous velocity is defined as

$$\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$

- Its direction is along the tangent to the path at each point of the path.



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Instantaneous velocity

- The components of the instantaneous velocity are

$$v_x = \frac{\mathrm{d}x}{\mathrm{d}t}, \quad v_y = \frac{\mathrm{d}y}{\mathrm{d}t} \text{ and } v_z = \frac{\mathrm{d}z}{\mathrm{d}t}$$

- In terms of unit vectors $\vec{v} = \frac{\mathrm{d}\vec{r}}{\mathrm{d}t} = \frac{\mathrm{d}x}{\mathrm{d}t}\hat{i} + \frac{\mathrm{d}y}{\mathrm{d}t}\hat{j} + \frac{\mathrm{d}z}{\mathrm{d}t}\hat{k}$
- The magnitude of the instantaneous velocity is the speed of the object

$$v = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

Motion in *x*-*y* plane

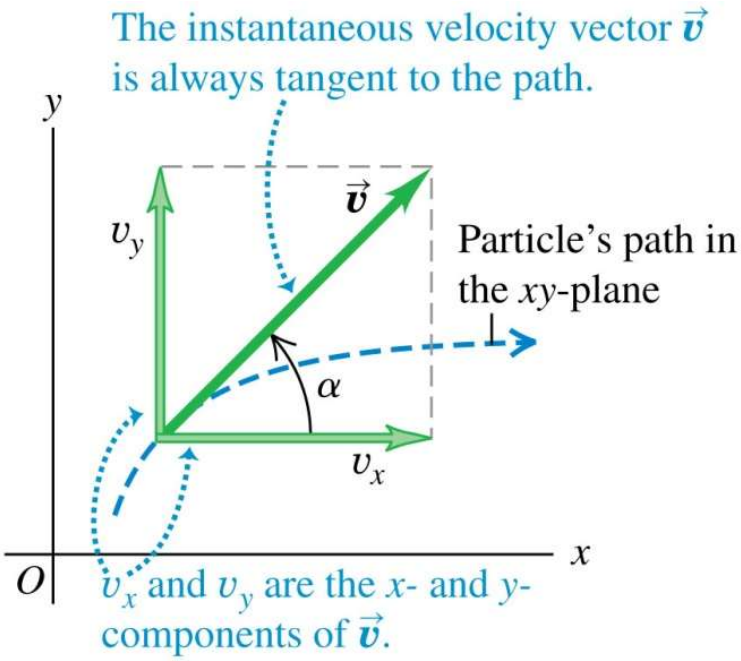
- When a particle moves in the x-y plane (see below figure) then speed is

$$v = \sqrt{v_x^2 + v_y^2}$$

- The direction of the velocity vector is terms of unit vectors as

$$\alpha = \tan^{-1} \frac{v_y}{v_x}$$

- The velocity is always tangent to the particle's path



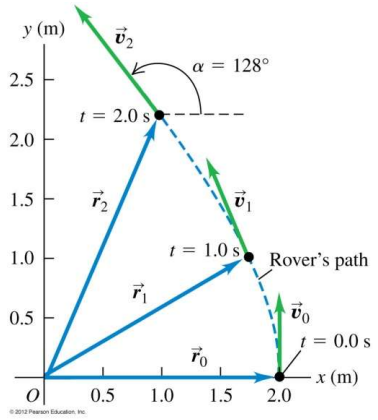
Example 9

A robotic vehicle, or rover, is exploring the surface of Mars. The landing craft is the origin of the co-ordinates, and the surrounding Martian surface lies in the *x*-*y* plane. The rover, which we represent as a point, has *x*- and *y* co-ordinates that vary with time :

$x = 2.0 - 0.25t^2$

$y = 1.0t + 0.025t^3$

a) Find the rover's co-ordinates and its distance from the lander at $t = 2.0$ s.
b) Find the rover's displacement and average velocity vectors during the time interval from $t = 0.0$ s to $t = 2.0$ s.
c) Derive a general expression for the rover's instantaneous velocity vector.
d) Express the instantaneous velocity at $t = 2.0$ s in component form and also in terms of magnitude and direction.

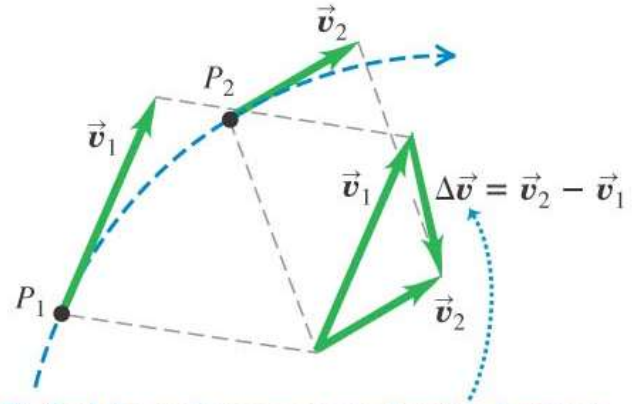


Average acceleration

- The average acceleration is the change in instantaneous velocity $\Delta\vec{v}$ divided by the time interval Δt.

$$\vec{a}_{av} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} = \frac{\Delta\vec{v}}{\Delta t}$$

- Its direction is in the direction of $\Delta\vec{v}$.



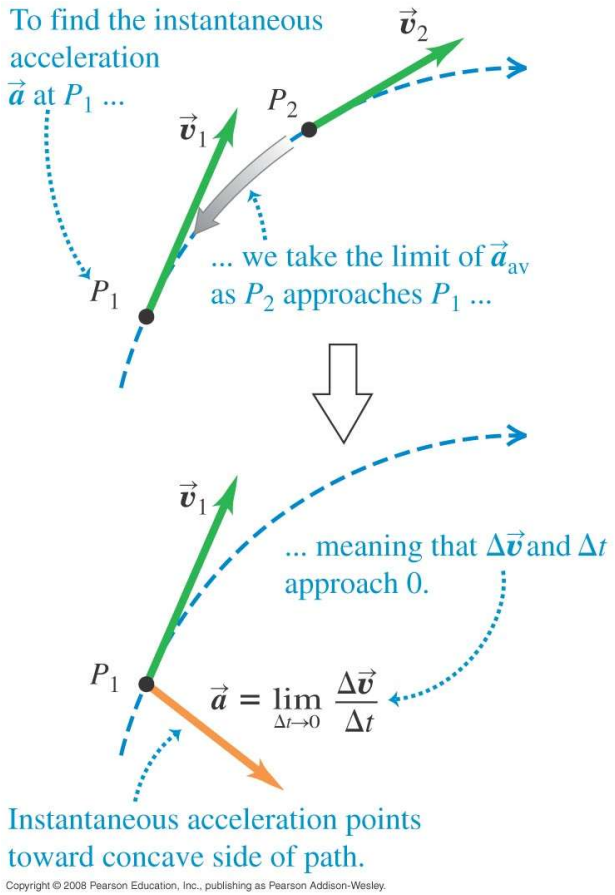
To find the car's average acceleration between P_1 and P_2, we first find the change in velocity $\Delta\vec{v}$ by subtracting $\vec{v}_1$ from $\vec{v}_2$. (Notice that $\vec{v}_1 + \Delta\vec{v} = \vec{v}_2$.)

Instantaneous acceleration

- The instantaneous acceleration is defined as :

$$\vec{a} = \lim_{\Delta t \to 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt}$$

- Its direction is towards the concave side of the path.



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Instantaneous acceleration

- The components of the instantaneous acceleration are

$$a_x = \frac{\mathrm{d}v_x}{\mathrm{d}t} = \frac{\mathrm{d}^2 x}{\mathrm{d}t^2}, \quad a_y = \frac{\mathrm{d}v_y}{\mathrm{d}t} = \frac{\mathrm{d}^2 y}{\mathrm{d}t^2}, \quad a_z = \frac{\mathrm{d}v_z}{\mathrm{d}t} = \frac{d^2 z}{dt^2}$$

- We can also write the acceleration vector in terms of unit vectors as

$$\vec{a} = \frac{\mathrm{d}\vec{v}}{\mathrm{d}t} = \frac{\mathrm{d}v_x}{\mathrm{d}t}\hat{i} + \frac{\mathrm{d}v_y}{\mathrm{d}t}\hat{j} + \frac{\mathrm{d}v_z}{\mathrm{d}t}\hat{k}$$

$$= \frac{\mathrm{d}^2\vec{r}}{\mathrm{d}t^2} = \frac{\mathrm{d}^2 x}{\mathrm{d}t^2}\hat{i} + \frac{\mathrm{d}^2 y}{\mathrm{d}t^2}\hat{j} + \frac{\mathrm{d}^2 z}{\mathrm{d}t^2}\hat{k}$$

Example 10

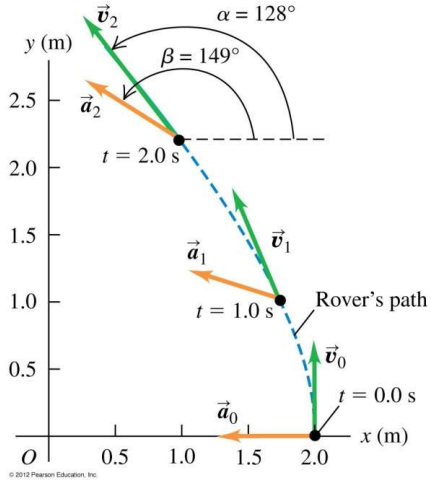
A robotic vehicle, or rover, is exploring the surface of Mars. The landing craft is the origin of the co-ordinates, and the surrounding Martian surface lies in the *x*-*y* plane. The rover, which we represent as a point, has *x*- and *y* co-ordinates that vary with time :

$$x = 2.0 - 0.25t^2$$

$$y = 1.0t + 0.025t^3$$

a) Find the components of the average acceleration during the time interval from $t = 0.0$ s to $t = 2.0$ s.

b) Find the instantaneous acceleration at $t = 2.0$ s.

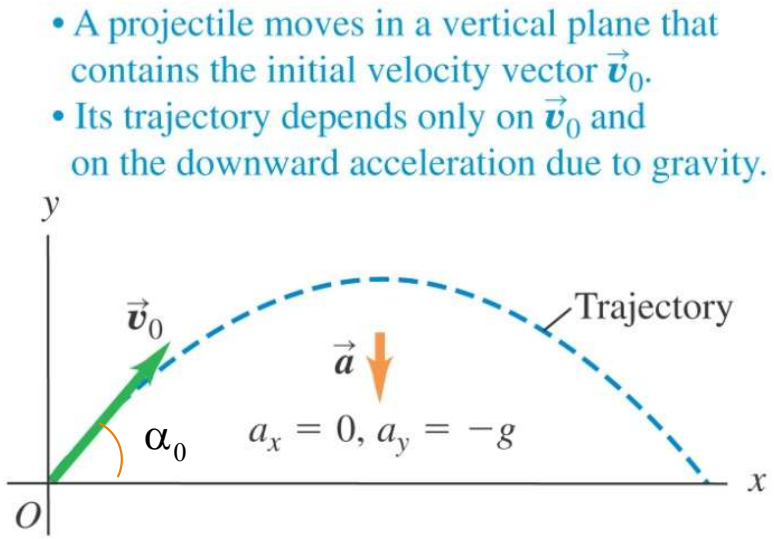


Projectile motion

- A projectile is any body that is given an initial velocity and then follows a path determined entirely by the effects of gravity and air resistance.
- We will set up equations for a projectile motion assuming no air resistance.
- The initial velocity v_0 can be resolved into the x and y components in terms of the angle α_0, i.e.

$$v_{0x} = v_0 \cos\alpha_0$$

$$v_{0y} = v_0 \sin\alpha_0$$



Projectile motion

- If there is no air resistance, the horizontal velocity $v_0 \cos\alpha_0$ is constant.
- The horizontal distance travelled in time t is $x = (v_0 \cos\alpha_0)\, t$.
- Because of gravity, the vertical velocity varies as $v_y = (v_0 \sin\alpha_0) - gt$
- The vertical distance travelled is in time t is $y = (v_0 \sin\alpha_0)t - \frac{1}{2} gt^2$
- Eliminating t from the equation for x and y, we have,

  $$y = (\tan\alpha_0)\, x - \frac{1}{2v_0^2 \cos^2\alpha_0} gx^2$$
- This shows that the object moves in a parabolic path.

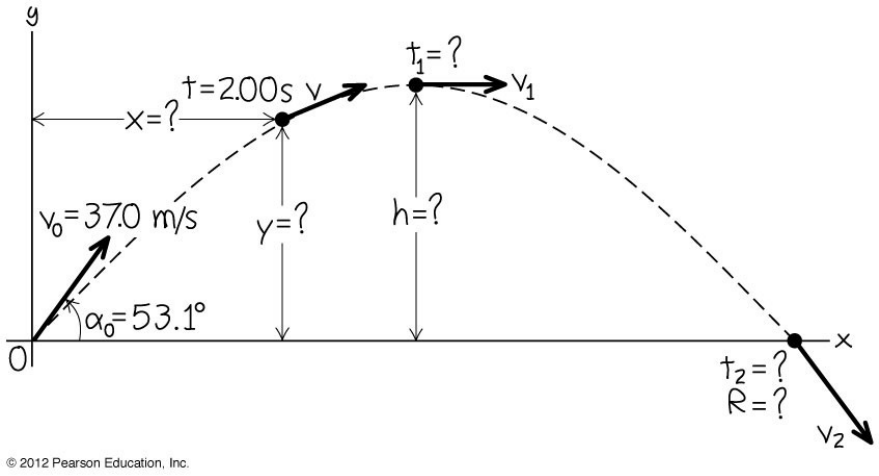
Example 11

A batter hits a baseball so that it leaves the bat at speed $v_0 = 37.0$ m/s at an angle $\alpha_0 = 53.1^o$.

a) Find the position of the ball and its velocity at $t = 2.00$ s.

b) Find the time when the ball reaches the highest point of its flight and its height h, at this time.

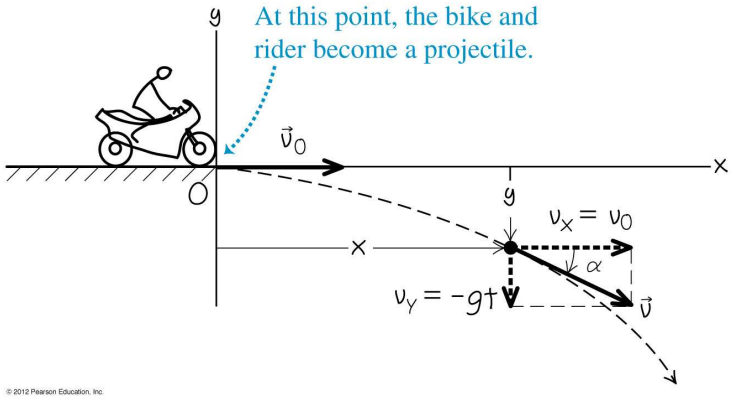
c) Find the horizontal range R which is the maximum horizontal distance travelled by the ball.



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Example 12

A motorcycle stunt rider rides off the edge of a cliff. Just at the edge his velocity is horizontal, with magnitude 9.0 m/s. Find the motorcycle's (a) position (b) distance from the edge of the cliff and (c) velocity 0.5 s after it leaves the edge of the cliff.

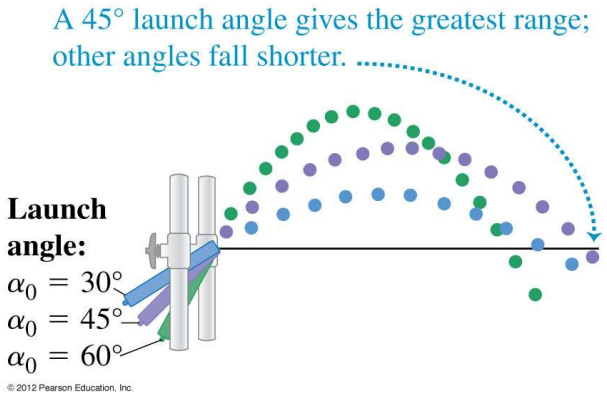


Example 13

Find the maximum height h and horizontal range R of a projectile launched with speed v_0 at an initial angle α_0 between 0° and 90° .

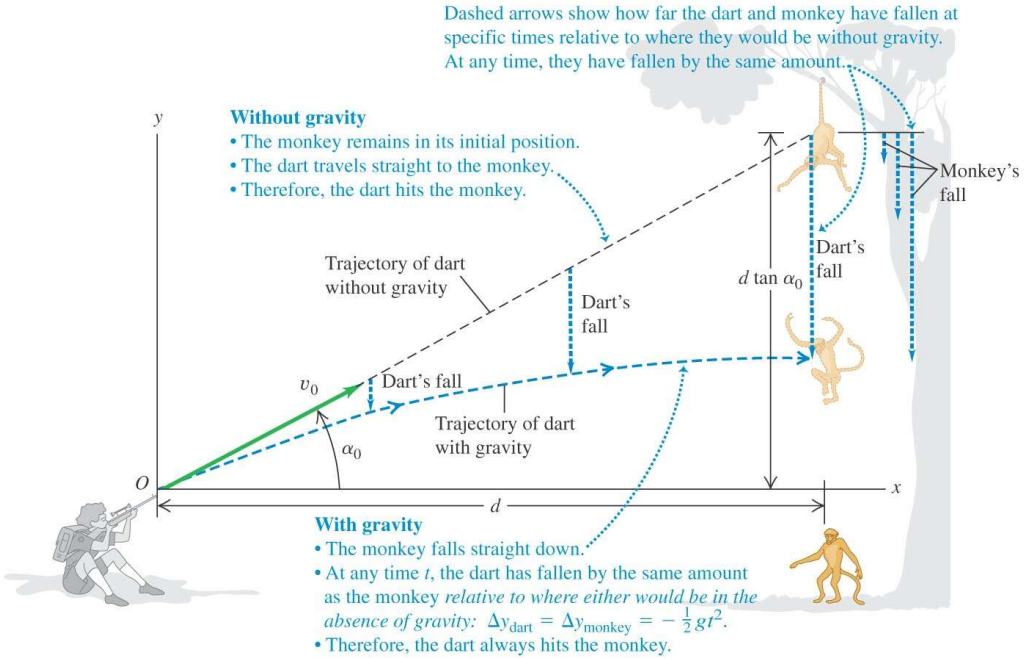
a) For a given $v_{0,}$ what value of α_0 gives maximum height?

b) What value of α_0 gives maximum horizontal range?



Example 14

A monkey escapes from the zoo and climbs a tree. After failing to entice the monkey down, the zookeeper fires a tranquilizer dart directly at the monkey (see below figure). The monkey lets go at the instant the dart leaves the gun. Show that the dart will always hit the monkey, provided that the dart reaches the monkey before the monkey hits the ground and runs away.

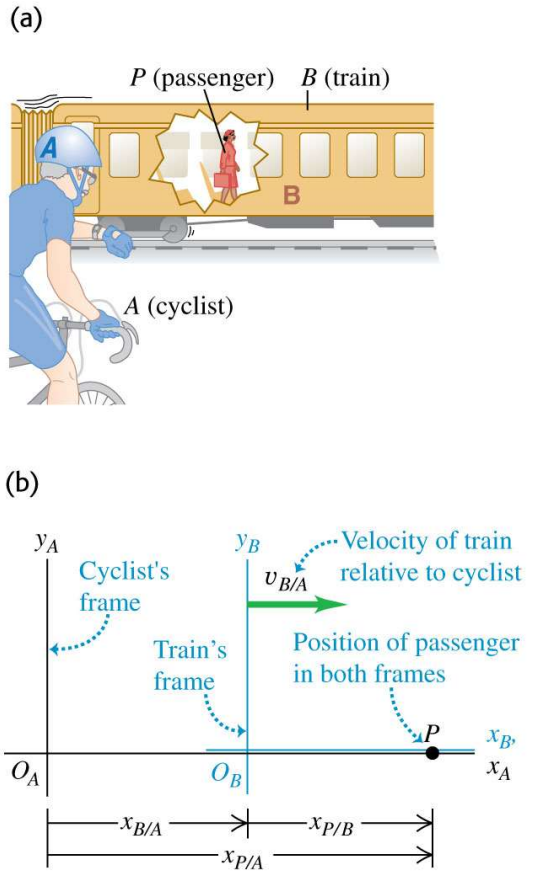


Relative velocity

- The velocity of a moving body seen by an observer is called relative velocity.
- The observer may be stationary or moving.
- To study relative velocity we need a frame of reference.
- A frame of reference is a coordinate system plus a time scale.

Relative velocity in 1D

- In figure (b), the x position of P relative to A is
$$\vec{x}_{\mathrm{P/A}} = \vec{x}_{\mathrm{P/B}} + \vec{x}_{\mathrm{B/A}}$$
- Imagine the subscripts as a fraction and split it as P/A = P/B × B/A.
- The velocity of P relative to A is obtained by differentiation.
$$\frac{\mathrm{d}\vec{x}_{\mathrm{P/A}}}{\mathrm{d}t} = \frac{\mathrm{d}\vec{x}_{\mathrm{P/B}}}{\mathrm{d}t} + \frac{\mathrm{d}\vec{x}_{\mathrm{B/A}}}{\mathrm{d}t}$$
$$\vec{v}_{\mathrm{P/A-x}} = \vec{v}_{\mathrm{P/B-x}} + \vec{v}_{\mathrm{B/A-x}}$$

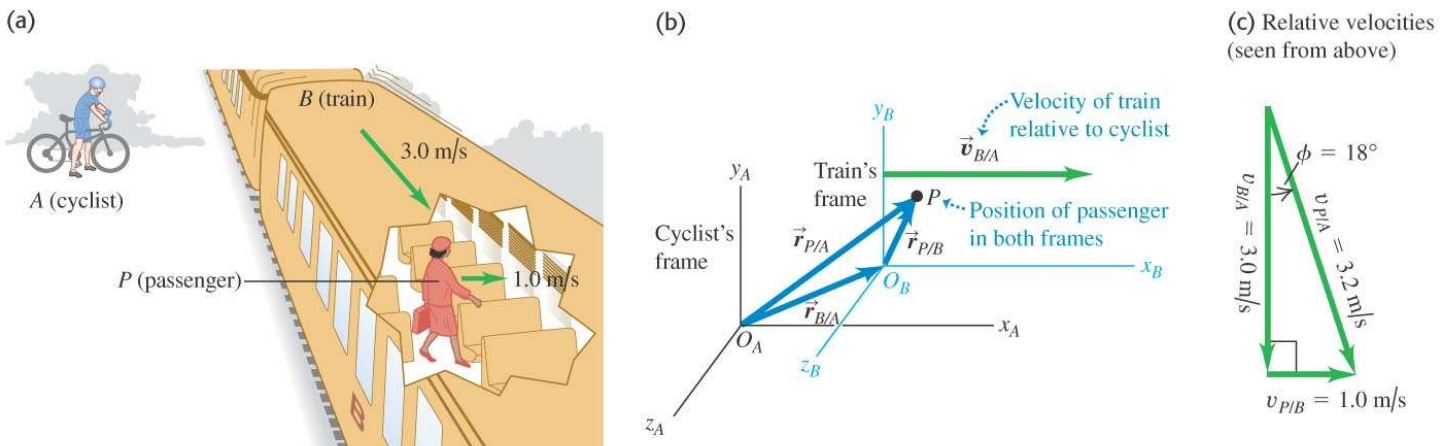


Relative velocity in 2D and 3D

- In 2D and 3D, the velocity of P relative to A is obtained by differentiating the position vector $\vec{r}_{P/A} = \vec{r}_{P/B} + \vec{r}_{B/A}$

$$\frac{d\vec{r}_{P/A}}{dt} = \frac{d\vec{r}_{P/B}}{dt} + \frac{d\vec{r}_{B/A}}{dt}$$

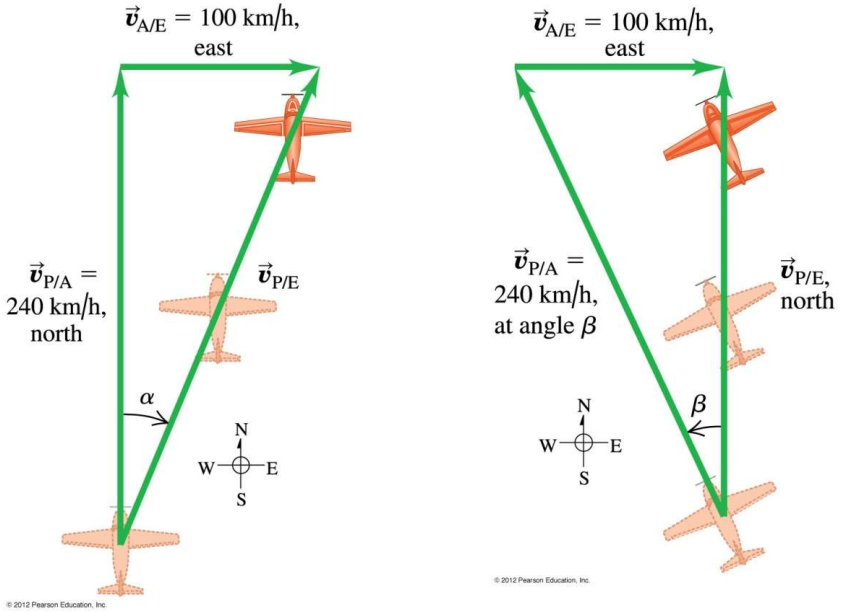
$$\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A}$$



Example 15

a) An air-plane's compass indicates that it is headed due north and its airspeed indicator shows that it is moving through the air at 240 km/h. If there is a 100-km/h wind from west to east, what is the velocity of the airplane relative to the earth?

b) With the same wind and airspeed, in what direction should the airplane head to travel due north? What will be the velocity of the airplane relative to the earth?



Motion with a non-uniform acceleration

- The following equations are useful in the case of non-uniform acceleration.

$$\vec{a} = \frac{\mathrm{d}\vec{v}}{\mathrm{d}t}$$

Integrating we get $\vec{v} = \vec{v}_0 + \int_0^t \vec{a}\ \mathrm{d}t$

$$\vec{v} = \frac{\mathrm{d}\vec{r}}{\mathrm{d}t}$$

Integrating we get $\vec{r} = \vec{r}_0 + \int_0^t \vec{v}\ \mathrm{dt}$

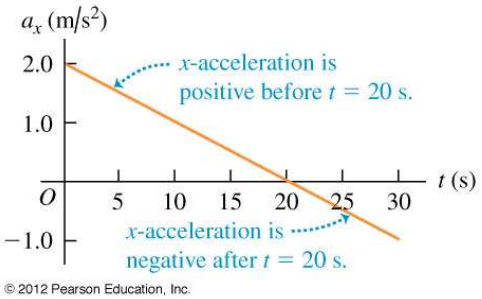
Example 16

Sally is driving along a straight highway. At $t = 0$, when she is moving at 10 m/s in the positive x-direction, she passes a sign-post at $x = 50$ m. Her x-acceleration as a function of time is $a_x = 2.0 - 0.10t$ m/s^2

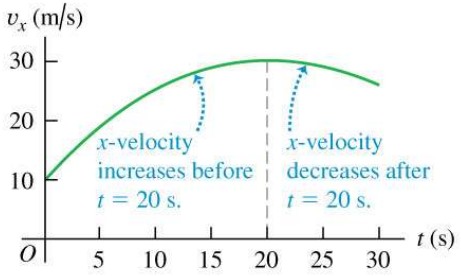
a) Find her x-velocity v_x and position x as functions of time.

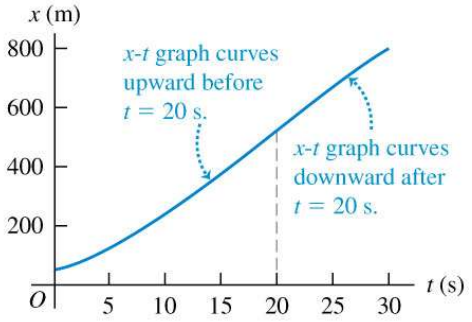
b) When is her x-velocity maximum?

c) What is that maximum x-velocity and where is it reached?



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End of chapter